

Is fermion parity protected by topology under noise, and how does the chain length L matter?

Seminar notes (Block 4 – the professor’s questions)

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Abstract

The professor asked whether the fermion parity of the Kitaev-chain simulation is “protected by topology,” and how the chain length L enters. We answer both in the noisy setting built in Week 9, applying Qiskit’s single-qubit decoherence channels to the topological ground state. The short answers: (Q1) what is conserved is conserved by *symmetry*, not topology — phase damping keeps $\langle \hat{P} \rangle$ exactly fixed because it is parity-preserving, while the genuinely topological object, the non-local edge string, is not noise-immune at any fixed L ; (Q2) the chain length acts in *opposite* directions — the intrinsic parity gap is protected exponentially, $\Delta_P(L) \sim e^{-L/\xi}$, but the vulnerability to T_1 poisoning and to depolarizing contrast loss *grows* with L .

1 Background: symmetry versus topology

Under Jordan–Wigner the Kitaev chain is

$$H = -\frac{\mu}{2} \sum_j Z_j + \frac{t-\Delta}{2} \sum_j X_j X_{j+1} + \frac{t+\Delta}{2} \sum_j Y_j Y_{j+1}, \quad \hat{P} = \prod_j Z_j. \quad (1)$$

Every term has an even number of X/Y factors, so $[H, \hat{P}] = 0$ in *every* phase: parity conservation is a $\mathbb{Z}_2$ symmetry of the Hamiltonian, not a topological effect. What topology adds is that in the topological phase ($|\mu| < 2t$) the two Majorana ends fuse into one non-local fermion $f = \frac{1}{2}(\gamma_L + i\gamma_R)$ whose occupation is the parity bit; the even/odd ground states become degenerate with splitting (the *parity gap*)

$$\Delta_P(L) = |E_0^+ - E_0^-| \approx 2t(1 - \lambda^2)\lambda^L, \quad \lambda = \frac{|\mu|}{2t} = e^{-1/\xi}. \quad (2)$$

Because f joins the two ends, no *local* operator can read or flip the bit. The question for Block 4 is what survives once the device decoheres.

2 Which noise channels respect parity

A single-qubit channel matters for $\hat{P}$ only through how its Kraus operators commute with the local Z_j . Using Qiskit’s standard channels:

Channel (Qiskit)	Acts through	vs. $\hat{P} = \prod_j Z_j$	Effect on $\langle \hat{P} \rangle$
phase_damping_error (T_2)	Z_j	commutes (preserving)	exactly conserved
depolarizing_error	X_j, Y_j, Z_j	X, Y anticommute	shrinks $(1 - p)^L$ toward 0
amplitude_damping_error (T_1)	σ_j^-	violating, biased to $ 0\rangle$	leaks to odd sector

Z_j commutes with $\hat{P}$, so dephasing leaves $\langle \hat{P} \rangle$ untouched. X_j, Y_j and the lowering operator $\sigma^- = \frac{X + iY}{2}$ anticommute with Z_j , so a single such event flips the global parity. Amplitude damping is additionally *biased*: it preferentially sends $|1\rangle \rightarrow |0\rangle$. In the repository Jordan–Wigner convention $c_j^\dagger c_j = (I + Z_j)/2$, so $|0\rangle$ is the occupied local state; the essential point is that the channel changes the local Z_j eigenvalue and can therefore leak the protected even state into the odd sector rather than scrambling it symmetrically.

Connection to Week 9. The realistic `thermal_relaxation_error` (T_1, T_2, τ) used in the Week 9 gate-noise study is exactly the composition of these two pieces: its T_1 part is the *parity-violating* amplitude damping, its T_2 part is the *parity-preserving* dephasing. So the two questions below dissect the same physical channel that drives the circuit-level model.

3 Q1: is the parity protected by topology, under noise?

We prepare the exact even-parity ground state at $\mu = t$ (topological), apply each channel at increasing strength p , and read off $\langle \hat{P} \rangle$ and the edge string (Fig. 1). Two different objects behave very differently.

The conserved quantity $\langle \hat{P} \rangle$ — protected by *symmetry*. Under phase damping $\langle \hat{P} \rangle$ stays *exactly* 1 (flat to machine precision), because the Kraus operators commute with $\hat{P}$. But this is the $\mathbb{Z}_2$ symmetry, which holds in every phase — it is *not* topology. Both parity-violating channels break it: depolarizing drives $\langle \hat{P} \rangle \rightarrow 0$ as $(1 - p)^L$, and amplitude damping leaks into the odd sector ($\langle \hat{P} \rangle$ falling, but staying above depolarizing because the bias pulls back toward the $\hat{P} = +1$ vacuum).

The topological order parameter — *not* noise-immune. The non-local edge string decays under *every* channel, *including* the parity-preserving phase damping that leaves $\langle \hat{P} \rangle = 1$: dephasing destroys the cross-chain coherence the string relies on, so its contrast falls even while parity is perfectly conserved.

Answer. Parity conservation is protected against parity-preserving noise — but by *symmetry*, not topology — and is destroyed by parity-violating noise (T_1 poisoning, depolarizing). The genuinely *topological* object, the non-local edge string, is *not* protected against any noise at fixed L : conserving parity does not preserve the topological signal. Topology does not buy fixed- L noise immunity; where it enters is the chain-length scaling.

4 Q2: how does the chain length change this, under noise?

Length acts in two competing ways (Fig. 2).

- **Intrinsic protection improves with L .** Noiselessly, $\Delta_P(L) \sim e^{-L/\xi}$ in the topological phase ($\mu = t$): on a semilog axis a straight line of slope $\ln(|\mu|/2t)$, falling from 0.41 at $L = 2$ to 5.9×10^{-3} at $L = 8$. In the trivial phase ($\mu = 3t$) the gap is flat at $\mathcal{O}(1)$ — there is no degeneracy to protect.
- **Vulnerability worsens with L .** At fixed channel strength $\gamma = p = 0.1$, the odd-sector leakage under amplitude damping grows from 0.03 to 0.21 (more sites that can lose a fermion, $\sim 1 - (1 - \gamma)^L$), and the edge-string loss under depolarizing grows from 0.19 to 0.57 (a longer non-local string accumulates more single-qubit contrast loss).

Parity (symmetry) survives dephasing; the edge string (topology) does not ($L = 6, \mu = 1t$)

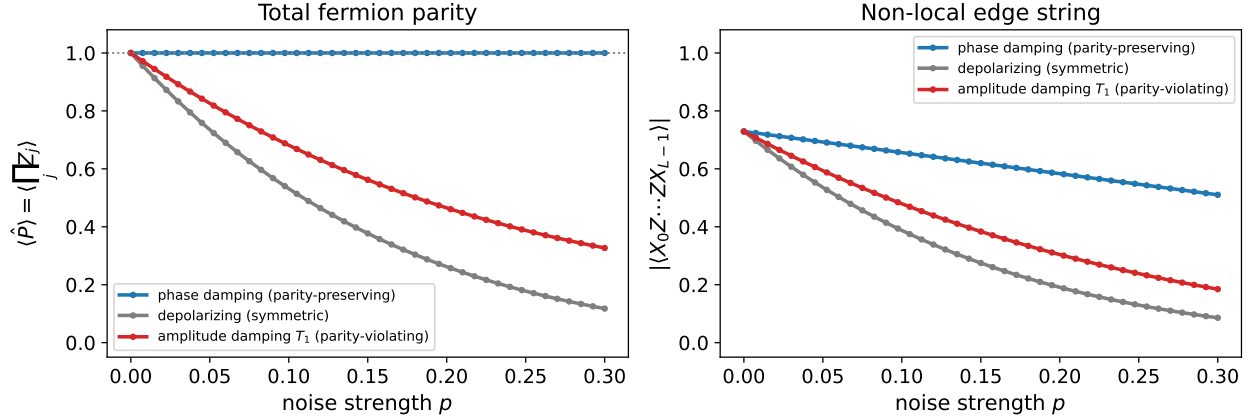


Figure 1: Topological state at $L = 6, \mu = t$, Qiskit single-qubit channels. Left: $\langle \hat{P} \rangle$ stays exactly 1 under phase damping (symmetry), collapses under depolarizing, and leaks (stays biased high) under amplitude damping. Right: the non-local edge string decays under *all* channels — including the parity-preserving phase damping — so parity conservation does not protect the topological signal.

Answer. There is no monotone “longer is better.” The topological degeneracy is exponentially better protected at large L , but the noisy device loses parity and contrast faster at large L . The two trends oppose, defining an optimal working length — the direct analogue, in the chain-length axis, of the optimal ansatz depth found in Week 9.

5 Summary

- Parity *conservation* is a $\mathbb{Z}_2$ symmetry, not topology; it survives parity-preserving (T_2) noise and is broken by parity-violating (T_1 , depolarizing) noise.
- The *topological* object (the non-local edge string / the degeneracy) is not noise-immune at fixed L — every channel erodes it.
- Topology’s role is in the L -scaling: intrinsic protection improves as $e^{-L/\xi}$ while the noisy device degrades with L , giving an optimal length — the chain-length counterpart of the Week 9 optimal-depth result.
- These three axes — noise strength, ansatz depth (Week 9), and chain length — are the knobs that decide whether a NISQ measurement of the Kitaev chain can still resolve the topological phase.

Chain length: opposite effects under noise

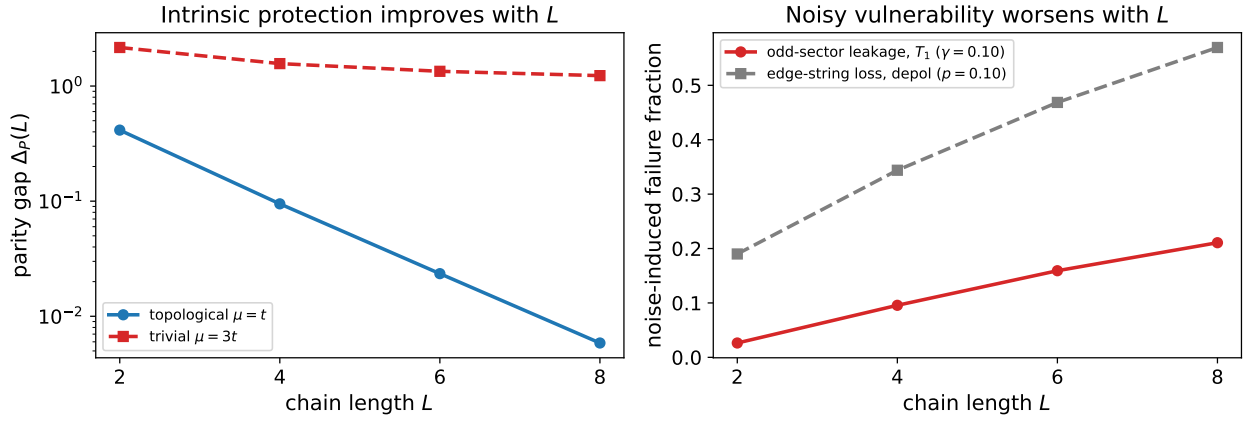


Figure 2: Left: noiseless parity gap $\Delta_P(L)$ — exponential decay in the topological phase (intrinsic protection improving with L) versus the flat $\mathcal{O}(1)$ trivial gap. Right: at fixed noise $\gamma = p = 0.1$, odd-sector leakage under T_1 and edge-string loss under depolarizing both grow with L . Longer chains are intrinsically better protected but noisier to operate.